15 Hypothesis testing, significance testing, confidence intervals and power

Subject content	Additional information
15.1 Use confidence intervals for the mean using z or t as appropriate, interpreting results in practical contexts.	Students are expected to use the t-distribution when the sample size is small ($n < 30$) and the population standard deviation, σ, is unknown. Students should be familiar with the notation $\bar{x}$ and s^2 as statistics evaluated from a sample for respective estimates of μ and σ^2. Students should know that the $(n-1)$ divisor is used when evaluating s^2. Students will be expected to learn the formulae for constructing a confidence interval for a population mean: $\bar{x} \pm (z \text{ or } t) \times (\text{standard error})$
15.2 Know that a change in sample size will affect the width of a confidence interval.	Students will be expected to be able to find the sample size required in order to obtain a confidence interval with a specified width.
15.3 Evaluate the strength of conclusions and misreporting of findings from hypothesis tests, including the calculation and importance of the power of a hypothesis test.	See 15.6
15.4 Know that sample size can be changed to potentially elicit appropriate evidence in a hypothesis test.	